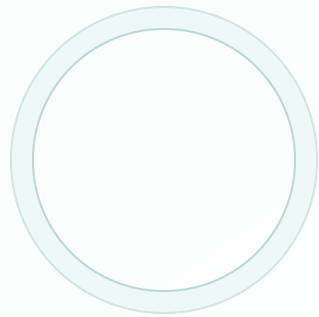




# AI Dental Scribe

An ambient documentation assistant that understands teeth, surfaces, periodontal findings, and dental procedure language - while leaving every clinical decision with the dentist.

|              |                                                             |
|--------------|-------------------------------------------------------------|
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| STATUS       | Portfolio / Concept PRD                                     |
| PRIMARY USER | General dentist                                             |

|                                                                           |                                                                  |                                                              |
|---------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|
| <b>1-2 hours/day</b><br>Documentation burden to validate during discovery | <b>&lt;1%</b><br>Proposed maximum hallucination rate for release | <b>5 practices</b><br>Proposed pilot across varied workflows |
|---------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------|

**Product decision**  
Build a dental-specific drafting layer, not an autonomous clinician. The system may listen with explicit consent, structure facts, and prepare a note; it may not diagnose, plan treatment, submit codes, or commit a clinical record without human review and sign-off.

# Dentistry creates a harder documentation problem than generic scribing

A clinically fluent note must preserve what happened chairside, identify exactly where it happened, and avoid turning an uncertain utterance into a recorded fact.

## Problem

Dentists often finish notes between patients or after clinic. The working discovery hypothesis is that documentation consumes 1-2 hours per day for many clinicians. Templates speed entry, but they also encourage copy-forward text and still require manual tooth, surface, material, anesthetic, and finding detail.

Generic medical scribes miss dentistry's compact vocabulary and location logic. A single phrase may encode a tooth, one or more surfaces, a finding, a procedure, material, anesthesia, and follow-up. Periodontal charting adds six-site measurements, recession, bleeding, mobility, and furcation data.

## Why dental specificity matters

- **Notation collision:** FDI 12 is the maxillary right lateral incisor; Universal #12 is the maxillary left first premolar.
- **Surface precision:** "MOD composite on 19" cannot become "filling on tooth 19."
- **Negation:** "No swelling or mobility" must not be recorded as a positive finding.
- **Perio structure:** 4-3-4 at MB-B-DB is not free text; site order matters.
- **Procedure language:** CDT candidates require the current licensed code set and clinician confirmation.[1]

## GOALS

### Reduce clerical work

Cut median active documentation time per encounter by at least 50% in the pilot.

### Increase same-day completion

Make review and signature possible before the clinician leaves the operatory.

### Earn clinical trust

Make every extracted entity traceable, editable, and confidence-aware.

## NON-GOALS

### No autonomous clinical judgment

No independent diagnosis, radiographic interpretation, prognosis, or treatment planning. The tool documents clinician-stated content only.

### No unreviewed administrative action

No billing-code submission, claim creation, prescription, order, referral, or chart finalization without authorized human review.

**Scope assumption:** U.S. general dentistry first; English-language adult encounters; a practice-selected Universal or FDI notation mode. Pediatric/mixed-dentition support is evaluated before being enabled.

# The dentist signs; the clinical team shapes the workflow

## General dentist

**Primary job:** finish a defensible clinical note without breaking patient focus.

**Pain:** fragmented dictation, template cleanup, late notes, wrong tooth/surface risk.

**Trust need:** evidence-linked facts and fast correction.

## Dental hygienist

**Primary job:** document history updates, periodontal findings, preventive care, and education.

**Pain:** dense measurement entry and switching between chart and narrative.

**Trust need:** site-level accuracy and no invented normal findings.

## Office manager

**Primary job:** keep charts complete, workflows compliant, and integrations reliable.

**Pain:** unsigned notes, inconsistent templates, access and audit questions.

**Trust need:** consent visibility, role controls, and completion reporting.

## USER STORIES

1. **As a dentist**, I want the patient-consent state visible before capture so that I never record an encounter without permission.
2. **As a dentist**, I want spoken tooth numbers resolved using my practice's notation so that "12" cannot map to the wrong tooth.
3. **As a dentist**, I want surfaces, materials, anesthesia, and complications separated into editable fields so that I can verify the procedure note quickly.
4. **As a dentist**, I want low-confidence phrases highlighted beside the transcript evidence so that I know where to focus my review.

5. **As a hygienist**, I want six-site periodontal measurements and bleeding/recession findings mapped to the correct tooth and site so that I do not re-enter the chart.
6. **As an office manager**, I want role-based access and a complete edit/signature trail so that I can support compliance reviews.
7. **As a dentist**, I want a SOAP draft in my preferred template so that the output reads like my practice's record, not a generic summary.
8. **As a dentist**, I want procedure-code candidates clearly marked as suggestions so that only I decide what is documented or billed.

## CORE JOB-TO-BE-DONE

**When I complete an encounter**, help me turn the chairside conversation and my spoken findings into a structured, dental-specific draft that I can verify and sign in minutes - without adding facts I did not say.

## PRIMARY JOURNEY

### 1. Prepare

Confirm patient, notation, template, and permitted context.

### 2. Consent

Explain capture; record explicit accept or decline.

### 3. Capture

Stream audio; show pause/stop state at all times.

### 4. Review

Verify entities, uncertainties, and SOAP draft.

### 5. Sign

Clinician approves; final note writes to the record.

# The workflow keeps AI drafting behind consent and review gates

| PRIORITY | CAPABILITY                        | REQUIREMENT AND ACCEPTANCE SIGNAL                                                                                                                                                                                 |
|----------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P0       | <b>Consent-controlled capture</b> | Capture is disabled until staff records explicit consent. UI shows recording state; pause, stop, decline, and revoke are one action away. Consent event includes user, patient, time, script/version, and method. |
| P0       | <b>Real-time draft</b>            | Transcript and SOAP draft update during the encounter without silently committing to the chart. Dentist can mark a phrase "not for note."                                                                         |
| P0       | <b>Dental entity extraction</b>   | Extract tooth identifier, dentition, surface, condition/finding, negation, procedure, material, anesthetic, measurement, medication, consent, complication, and follow-up.                                        |
| P0       | <b>Notation awareness</b>         | Practice selects Universal or FDI; output stores canonical anatomy plus source notation. Ambiguous spoken numbers trigger a confirm step, never silent conversion.                                                |
| P0       | <b>Human review and sign-off</b>  | Draft remains unsigned and visually distinct. Only an authorized clinician can accept, edit, and sign. All AI-originated text and subsequent edits are auditable.                                                 |
| P0       | <b>Uncertainty display</b>        | Low-confidence spans are highlighted and linked to transcript timestamps. A note cannot be signed while required high-risk fields remain unresolved.                                                              |
| P1       | <b>Perio-aware structure</b>      | Capture six sites per tooth, bleeding on probing, recession, mobility, and furcation without inventing unspoken "normal" values.                                                                                  |
| P1       | <b>Template personalization</b>   | Support practice-approved exam, restorative, crown, extraction, endodontic, and hygiene templates with locked required fields.                                                                                    |

## Integration contract

- **Read, minimum necessary:** patient/appointment identifier, provider, active allergies/medications, existing tooth chart, and selected note template.
- **Write, staged:** draft to review queue first; signed note only after clinician action; code candidates remain unsubmitted.
- **Adapter pattern:** normalize vendor fields behind a common schema. Open Dental documents REST resources for patients, perio measures, procedure codes/logs, notes, and security logs.[2]
- **Dentrix Ascend path:** treat partner access, permissions, note-write capability, and sandbox availability as discovery gates before commitment.

**Performance target:** first readable draft within 10 seconds of stopping capture; autosave of review edits within 2 seconds; no signed-note write without a successful user-authenticated confirmation.

### Draft example - restorative encounter

**S:** Patient reports sensitivity to cold, lower left, two weeks. No spontaneous pain.

**O:** Universal #19, occlusal caries; no swelling. Local anesthetic documented. MOD preparation and composite placement stated.

**A:** Clinician-stated assessment only. **Low confidence:** "reversible pulpitis?"

**P:** Post-op instructions discussed; monitor symptoms. CDT candidate shown separately; not submitted.

# Model quality is a release gate, not a dashboard after launch

Evaluation compares the generated draft with a dentist-corrected reference note and the source transcript. Every metric is sliced by procedure type, notation system, speaker accent, background noise, and practice. Aggregate scores cannot hide a weak safety-critical slice.

## <1%

### Hallucination rate

Unsupported atomic clinical claims divided by all generated atomic clinical claims. A claim is unsupported when it appears in the draft but not in the encounter, permitted chart context, or clinician-entered data.

## ≤2%

### Critical omission rate

Critical gold facts absent from the draft divided by all critical gold facts. Zero tolerated high-severity misses in the release set for allergy, adverse event, consent, wrong-tooth prevention, or stated complication.

## CLINICAL ENTITY ACCURACY

| ENTITY                         | SCORING RULE                                                                             | FAILURE EXAMPLE                                                              |
|--------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Tooth and dentition</b>     | Exact match after notation normalization; report Universal and FDI separately.           | FDI 12 converted to Universal #12 rather than anatomical tooth identity.     |
| <b>Surface set</b>             | Exact set and per-surface precision/recall for M, D, O/I, B/F, L.                        | "MOD" reduced to "occlusal," or buccal mistaken for lingual.                 |
| <b>Finding and negation</b>    | Concept, anatomical site, assertion status, and temporality must all match.              | "No swelling" becomes "swelling," or historical pain becomes current.        |
| <b>Procedure and material</b>  | Exact procedure family plus linked tooth/surface/material; micro and macro F1.           | Core buildup attached to the wrong tooth; temporary crown recorded as final. |
| <b>CDT candidate</b>           | Top-1 exact code and top-3 recall against dentist-approved candidate; never auto-submit. | D2391 suggested when the documented surface count does not support it.       |
| <b>Perio measurement</b>       | Exact tooth + site + value; evaluate six-site completeness and impossible-value checks.  | 4-3-4 mapped to lingual instead of buccal sites.                             |
| <b>Anesthesia / medication</b> | Drug, concentration, amount, route/site, and response scored as linked attributes.       | Two carpules becomes one; epinephrine concentration is omitted.              |

## Metric hierarchy

**Safety gate first:** hallucinations, critical omissions, wrong tooth/surface, negation reversal. **Clinical utility second:** entity F1 and note completeness. **Style last:** formatting, preferred wording, and note length.

## ERROR SEVERITY

### Severe

Could alter care or record integrity: wrong tooth, invented diagnosis, omitted allergy/complication.

### Moderate

Requires meaningful correction: wrong surface, material, measurement, or procedure detail.

### Minor

No clinical meaning changed: grammar, section order, or stylistic preference.

# A dentist-annotated dataset makes regression measurable

## Planned evaluation set: minimum 240 encounters

| PROCEDURE STRATUM     | MINIMUM | KEY ENTITIES                      |
|-----------------------|---------|-----------------------------------|
| Exams / urgent visits | 40      | CC, HPI, findings, negation       |
| Direct restorations   | 40      | Tooth, surfaces, material         |
| Crowns / onlays       | 40      | Prep, material, shade, temp       |
| Extractions           | 40      | Tooth, anesthesia, complication   |
| Root canal therapy    | 40      | Canals, working length, irrigant  |
| Hygiene / periodontal | 40      | Six-site charting, BOP, recession |

### Dataset construction rules

- Use de-identified or properly authorized encounters under an approved data-governance process.
- Pair consented audio, verbatim transcript, AI draft, dentist-corrected note, and structured gold entities.
- Balance Universal and FDI notation examples; include spoken anatomy ("lower left first molar") as a cross-check.
- Represent accents, speech rates, masks, suction/high-speed noise, interruptions, multiple speakers, and silent procedures.
- Include difficult negatives and temporal cues: "no pain," "previous RCT," "consider crown next visit."
- Prevent patient or practice leakage across train, development, and test partitions.

### HUMAN EVALUATION RUBRIC

| DIMENSION                    | 1 = UNACCEPTABLE                            | 5 = READY TO SIGN                                         |
|------------------------------|---------------------------------------------|-----------------------------------------------------------|
| <b>Clinical fidelity</b>     | Meaning changed or facts invented           | All claims supported and correctly qualified              |
| <b>Critical completeness</b> | Safety-relevant facts missing               | All required findings, procedures, and follow-up present  |
| <b>Dental specificity</b>    | Generic summary; tooth/site links lost      | Correct notation, surfaces, measurements, and terminology |
| <b>Edit burden</b>           | Faster to rewrite                           | Only optional stylistic edits                             |
| <b>Safe uncertainty</b>      | Uncertainty hidden or resolved by invention | Ambiguity highlighted with useful evidence                |

### Inter-rater agreement

Two dentists independently annotate at least 20% of the frozen test set. Use Cohen's kappa for categorical entity labels and weighted kappa for the 1-5 rubric. Target  $\geq 0.80$ . Below 0.70 triggers guideline revision, re-training, and re-annotation before model conclusions are accepted.

### Adjudication

Disagreements go to a third clinician. The adjudication log records whether the root cause was ambiguous audio, unclear annotation guidance, terminology mapping, or model error. Annotation rules version with the dataset.

### REGRESSION POLICY FOR EVERY MODEL OR PROMPT CHANGE

1. Run the immutable safety set and complete 240+ encounter suite against the release candidate and current production baseline.
2. Block release if hallucination rate reaches 1%, critical omission exceeds 2%, any severe failure is introduced, or tooth/surface F1 drops by more than 1 percentage point.
3. Require slice-level review and written sign-off from product, clinical safety, engineering, and privacy/security owners.

# Safety is designed into storage, inference, editing, and escalation

## Guardrails

- **Explicit consent:** no recording until the patient accepts; declining never affects care. Practice config reflects applicable state recording law and approved script.
- **Minimum retention:** default to delete raw audio after transcription/quality checks; any longer period requires a documented purpose, policy, and customer control. HHS notes retained recordings may become part of a designated record set when used for decisions.[3]
- **HIPAA controls:** business associate agreements where applicable; role-based access, unique identity, encryption, audit controls, integrity protection, incident response, and risk assessment. HHS describes administrative, physical, and technical safeguards for ePHI.[4]
- **No training by default:** customer protected health information is excluded from model training unless a separate, compliant agreement explicitly authorizes it.
- **Uncertainty first:** highlight low-confidence spans; require confirmation for tooth, surface, diagnosis, procedure, anesthesia, allergy, or complication uncertainty.
- **Hard clinical boundary:** block novel diagnosis or treatment-plan generation. If the clinician did not state it, the draft cannot present it as fact.
- **Audit trail:** preserve model version, prompt/template version, timestamps, source spans, user edits, approver, and final-sign event.

## Escalation path

1. **Flag**  
System or reviewer marks unsupported, omitted, privacy-sensitive, or clinically ambiguous content.
2. **Contain**  
Keep note unsigned; suppress write-back; retain only the minimum evidence needed under incident policy.
3. **Review**  
Dentist resolves the chart; clinical safety triages severity; privacy/security joins when ePHI is implicated.
4. **Correct**  
Update rule, annotation guideline, model/prompt, or integration mapping; add a regression case.
5. **Release**  
Only after the full evaluation suite passes and accountable owners sign off.

## FIVE-PRACTICE PILOT

| PHASE                                | MODE              | WHAT HAPPENS                                                                                                    | EXIT GATE                                                   |
|--------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>0. Readiness</b><br>2 weeks       | Configuration     | Security review, BAA, consent script, role setup, templates, baseline time study, integration test.             | No critical privacy or write-path defects.                  |
| <b>1. Shadow</b><br>2 weeks          | Invisible draft   | With consent, generate notes outside the chart; dentists annotate errors after completing normal documentation. | Safety thresholds pass by practice and procedure slice.     |
| <b>2. Assisted</b><br>3 weeks        | Review only       | Dentists see and edit drafts; no automatic chart write. Collect edit distance and time-on-task.                 | ≥70% accepted with no severe release-gate failure.          |
| <b>3. Supervised live</b><br>4 weeks | Signed write-back | Authorized dentist signs before note writes to PMS. Daily incident review in week one, then weekly.             | Efficiency and trust targets met for two consecutive weeks. |

**Pilot mix:** recruit five practices across solo and multi-provider settings, at least one hygiene-heavy workflow, and both supported notation styles. Final mix depends on integration access and data-governance approval.

# Success requires both efficiency and clinical trust

| METRIC                     | PILOT TARGET    | DEFINITION                                                                                                        |
|----------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Documentation time</b>  | ≥50% reduction  | Median active time from encounter end to signed note, compared with each clinician's baseline.                    |
| <b>Same-day completion</b> | ≥95%            | Eligible encounters with a signed note before local midnight on date of service.                                  |
| <b>Draft acceptance</b>    | ≥80%            | Drafts used as the basis of the final note, excluding consent declines and technical failures.                    |
| <b>Edit distance</b>       | Downward trend  | Clinician insertions, deletions, and substitutions normalized by final note length; inspect by entity class.      |
| <b>Dentist NPS</b>         | ≥30 directional | Likelihood to recommend after supervised-live use; pair with interviews because five practices is a small sample. |
| <b>Safety gates</b>        | <1% / ≤2%       | Hallucination rate / critical omission rate, plus zero newly introduced severe failures.                          |

## Top risks and mitigations

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>High</b>   | <b>Wrong tooth or surface</b><br>Practice-level notation lock, anatomy cross-check, mandatory ambiguity confirmation.             |
| <b>High</b>   | <b>Unsupported clinical statement</b><br>Evidence linking, no-inference prompt policy, hard release gate, signed human review.    |
| <b>High</b>   | <b>Consent/privacy failure</b><br>Capture off by default, logged explicit consent, minimal retention, security incident playbook. |
| <b>Medium</b> | <b>CDT drift or licensing</b><br>Licensed annual code set, effective-date checks, suggestions only.[1]                            |
| <b>Medium</b> | <b>Operatory noise and accents</b><br>Noise-aware test slices, headset option, confidence flags, easy replay.                     |
| <b>Medium</b> | <b>PMS variability</b><br>Adapter architecture, permission matrix, sandbox spike, staged write-back.                              |

## Open questions

1. Which launch PMS offers the safest supported draft-note write path: Open Dental, Dentrrix Ascend, or another partner?
2. What raw-audio retention default is clinically useful while remaining privacy-minimal?
3. Which states and consent scripts are in first-market scope?
4. Should CDT suggestions ship in the pilot, or follow core note accuracy?
5. Which note sections and required fields vary most by procedure and practice?
6. How should pediatric mixed dentition and Palmer notation enter the roadmap?
7. What latency is acceptable during high-speed handpiece or suction noise?

**Go/no-go rule:** Do not trade a safety threshold for time savings. A pilot advances only when the model passes clinical gates and dentists demonstrate faster completion with sustained use.

# Decisions and assumptions keep the concept honest

## Key product decisions

1. **Dental schema before fluent prose.** Tooth, surface, negation, measurement, and procedure links are first-class outputs.
2. **Drafting before automation.** The clinician remains the only authority who signs the note or accepts a code.
3. **Evidence before scale.** Evaluation thresholds and an immutable regression set govern every release.
4. **Adapters before vendor lock-in.** A common clinical-note contract isolates PMS-specific fields and permissions.
5. **Privacy-minimal by default.** Explicit consent, short retention, role controls, and traceable edits are core behavior.

## Assumptions to validate

- The 1-2 hour daily documentation burden is material for target users.
- A 240-encounter frozen test set is large and varied enough for initial release decisions.
- Five practices can expose meaningful workflow variation for an early pilot.
- At least one target PMS supports a safe draft-note review and signed write path.
- Explicit consent can be made fast enough not to interrupt rooming or care.
- English-first performance can meet safety gates across common accents and noisy operatories.

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**Portfolio note:** This is a concept PRD. Pilot targets, timelines, dataset size, and integration choices are proposed product decisions to validate, not results from a launched system.